

# Aarab Nishchal

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## Technical Skills

**Languages:** JavaScript, TypeScript, Python, SQL  
**Web & Frameworks:** React, Next.js, Node.js, Express, Angular, Django, FastAPI, Tailwind CSS  
**Databases & Cloud:** PostgreSQL, MongoDB, Supabase, ChromaDB, Convex, AWS EC2  
**AI / ML & LLMs:** PyTorch, LangChain, Ollama, Groq, VAPI, Whisper, Gemini, OpenRouter, RNN/LSTM  
**Tools:** Git, GitHub, CI/CD, Jenkins, n8n, REST APIs, Postman, Jira

## Experience

**Unstop** Sep 2024 – Present  
**Software Developer Intern** Remote

- Integrated AI-driven **n8n automation workflows** across internal **REST API endpoints** and a **custom CMS**, adding white-label support and dynamic window-title configuration
- Built an AI workflow system to auto-generate and validate vendor assignments, cutting creation and validation time by **60–70%**
- Enhanced front-end UI components to surface the new workflow system, streamlining vendor-facing task flows
- Managed **CI/CD pipelines** via **Git** and **Jenkins**, maintaining consistent deployments across environments
- Tracked sprint progress and maintained technical documentation in **Jira** and **Confluence**

## Projects

**TestIQ** | *LangChain · ChromaDB · Ollama · Tree-sitter · Python* GitHub

- Built a CLI **RAG pipeline** using **LangChain** and **ChromaDB** to analyze codebases and auto-generate unit tests, achieving 60 passing tests across 4 development phases
- Engineered a self-correcting validation loop with **Tree-sitter** (AST parsing) and **Ollama** (local inference) to eliminate invalid test output before finalization

**VidyaMarg** | *Next.js · FastAPI · Groq (LLaMA 3.1 70B) · NetworkX · Supabase · React Flow* GitHub

- Developed a full-stack AI learning path generator integrating **LLaMA 3.1 70B** via **Groq**, producing structured goal-based roadmaps from user input
- Architected graph-based path rendering with **NetworkX** and **React Flow**, visualizing learning paths as interactive **DAGs** backed by **Supabase**

**Curely** | *VAPI · OpenAI Whisper · Gemini · Openrouter* GitHub

- Designed an end-to-end real-time voice AI assistant for medical Q&A using **VAPI** and **Whisper**, with **Gemini** handling response generation across the full voice pipeline
- Reduced inference cost to **\$0.08/min** through targeted model selection and request routing via **Openrouter**

**Ideascribe** | *Convex · Clerk · React · Blocknote · Edgestore* GitHub

- Engineered a note taking application for developers, integrating **BlockNote** for dynamic rich-text editing, **Convex** for rapid state synchronization, and **Edge Store** for scalable cloud asset management.

## Education

**B.Tech, Computer Science & Engineering** 7.2 / 10 (till 7th Sem)  
*Kalinga Institute of Industrial Technology* 2022 – 2026  
**Senior Secondary (Class XII)** 67%  
*BSEB, BMP-7, Katihar, Bihar* 2020 – 2021  
**Secondary (Class X)** 78%  
*CBSE, New Light Academy, Namchi, Sikkim* 2019 – 2020

## Positions of Responsibility

**Team Lead** – *College Project Team, KIIT* 2023 – 2024

- Led a 6-member team across 3 college projects, managing task allocation, code reviews, and delivery milestones

**Lead Graphic Designer & Social Media Manager** – *Fashion Society, KIIT* 2022 – 2024

- Directed visual identity and social media strategy for college fashion events across two semesters

**Volunteer** – *Youth Red Cross Society, KIIT* 2022 – 2023

- Led a city-wide cloth donation drive, coordinating outreach and logistics across multiple residential zones

## Certifications

- AWS Certified Cloud Practitioner** – Amazon Web Services
- Scientific Computing with Python** – freeCodeCamp
- AMCAT Certified** in Software Development & Database Management – Aspiring Minds
- Problem Solving Certificate** – HackerRank
- Keploy Fellowship Program** – Open-source API testing and developer advocacy